

Required Capstone Checkpoint 1.1: Evaluating When Retrieval Is Required

Directions

Select one of the provided capstone scenarios and use it as the foundation for this checkpoint. You will continue developing this same scenario throughout the program. Download this worksheet and save a copy to your device before you begin working. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Use this worksheet to:

- Analyze how an LLM performs without retrieval
- Document prompts and corresponding model outputs
- Identify limitations and patterns in model behavior
- Determine when retrieval is required
- Reflect on how retrieval quality affects reliability and user trust

1. Selected Scenario and Intended Use

Select your capstone scenario:

- ☐ Research Paper Navigator
- ☒ Wikipedia Retrieval Engine

Briefly describe the kinds of questions, tasks, or user interactions your system should support.

Wikipedia is a free encyclopedia maintained by volunteers. The system I intend to build will be based upon a subset of roughly 7,000 articles from Wikipedia's 7+ million articles in English focused on significant people, places, and topics. I expect the system to be able to answer general interest questions such as what specific famous people are best known for, what they achieved in their lives, the impact they had; what is distinctive about specific places geographically or historically, what events have taken place there, what significant people is it known for, or what topics are connected to it; and what is known about a specific topic, its meaning, its scope, its parameters. For all this information, whether about people, places, or topics, the system needs to be able to provide fact-based answers to questions backed up with reputable sources that can be cited upon request.

Questions can be about a specific Wikipedia article or multiple articles e.g. they may ask the system to compare significant people, places or topics. The system should handle a back-and-forth conversation with the user, building trust by answering factually, never making up answers, and being honest and transparent about the limitations of its knowledge. I am also interested in being able to find relationships between Wikipedia articles (e.g. which use wikilinks, RDF properties, or categories)

2. Prompt and Output Evidence

- Provide two prompts relevant to your scenario. Where possible, use prompts that test different types of information needs or tasks.
- Include the corresponding LLM outputs for each prompt.

Prompt 1 and Corresponding LLM Outputs

PROMPT: "According to the Wikipedia article on Calton, Glasgow, what was the life expectancy of males in 2008? List the major contributing factors."

OUTPUT: "According to the Wikipedia article on **Calton, Glasgow**, the **life expectancy** for males in 2008 was 54 years**.**

The article says this was linked to several major factors, including:

- **deprivation / poverty**
- **poor health**
- **high levels of alcohol-related harm**
- **drug misuse**
- **smoking**
- **unemployment**

If you want, I can also quote the exact passage from the article."

Prompt 2 and Corresponding LLM Outputs

PROMPT: "Compare the 85th and 95th Academy Awards. How many awards were presented in each year and who presented the Best Supporting Actor Award in each year?"

OUTPUT: "The 85th and 95th Academy Awards each presented **24 awards**."

For the **Best Supporting Actor** award:

- **85th Academy Awards (2013)**: presented by **Octavia Spencer and Christopher Plummer**
- **95th Academy Awards (2023)**: presented by **Dwayne Johnson and Emily Blunt**

If you want, I can also compare the winners for that category in those two years.”

3. Analysis of Model Limitations

- Identify key limitations in the model’s responses, such as hallucinations, knowledge gaps, stale information, inconsistency, vagueness, or missing information.
- Use specific prompt/output examples to support your analysis.

Key limitations and supporting prompt/output examples

I have found that LLMs have a tendency to **hallucinate** when they are unable to find training data pertaining to the question. To elicit hallucination, it may be enough to ask them a **nonsensical question**. In my experience some LLMs do not question the premise of questions posed to them. For **example**, when asked *“According to its Wikipedia article, what is huba juice and why is it so delicious? Please provide a detailed explanation”*, the model responded in kind with a creative but nonsensical answer (there is no such thing as huba juice and no Wikipedia article exists for it): *“According to the Wikipedia article, **huba juice** is **not a real food or beverage** in any ordinary sense—it’s a **made-up joke term** from a viral internet meme. Wikipedia describes it as a **nonsensical phrase** used humorously, rather than an actual juice product.*

As for **why it is “so delicious,”** the joke is that there is **no real factual reason**. The “deliciousness” is part of the absurdity: it’s presented as if everyone should know and agree that it’s amazing, even though the term itself doesn’t refer to anything concrete. In other words, the humor comes from the mismatch between the **grand claim** (“so delicious”) and the **meaningless or fictional nature** of the thing being described.

I should note that I’m not able to verify the exact current wording of the Wikipedia article from here, so I can’t quote it directly. If you want, I can help you find the article or summarize its likely framing more closely.”

Another way to elicit hallucination is to **ask for a quote** from a specific work. For **example**, when the LLM was asked: *“According to the Wikipedia article on Charlotte Brontë, in 1846 she submitted her first novel, The Professor, along with her sister Emily’s Wuthering Heights and her sister Anne’s Agnes Grey. How did she describe these works to her London publisher? Please quote from her letter to*

him.” It responded very creatively, taking its lead from the prompt, and mimicking the writing style of the period, as follows: ‘In her letter to her London publisher George Smith, Charlotte Brontë described the three manuscripts as:

“I beg to submit to your consideration the accompanying work, ‘The Professor’—also a novel by my sister Emily, and one by my sister Anne. If you should not deem it advisable to publish ‘The Professor,’ perhaps you may be disposed to print that of my sister Emily—if not that of my sister Anne. They are all 3 the productions of the same pen.”

This is the wording commonly quoted from her 1846 letter accompanying the manuscripts.’

While George Smith would indeed later become her editor, it was Henry Colburn to whom she wrote in 1846 to pitch *The Professor*, the Wikipedia article actually cited the following quote from the letter - “Three tales, each occupying a volume and capable of being published together or separately, as thought most advisable” - and a little historical research reveals that her letter to him actually began “Sir – I request permission to send for your inspection the M.S. of a work of fiction in 3 vols. It consists of three tales, each occupying a volume ...” (see [Wuthering Heights – A Tale that Grew in the Telling? | Vulpes Libris](#)).

It is also evident that LLMs are **inconsistent** in their responses to closely related questions. For **example**, in reply to the question “*What is huba juice and why is it so delicious? Please provide a detailed explanation*”, the LLM did not always hallucinate, as described above. Note the slight variation in the question, which this time was more general and does not instruct the LLM to examine the Wikipedia article for huba juice. Sometimes an LLM can be forthcoming about not knowing the answer or it may seek more information from the prompter e.g. ‘I’m not certain what “huba juice” refers to. It doesn’t match a widely recognized food or drink name I’m aware of, so I don’t want to guess and invent details.

If you meant one of these, I can help:

- ****Hibiscus juice**** — a tart, ruby-red drink made from hibiscus flowers
- ****Huba**** as a local or brand name — if you share the spelling, region, or a label/photo, I can identify it
- A ****fictional drink**** from a book/game/meme

If you want, send me:

1. where you heard the term,
2. the spelling on the package/menu,
3. or a photo,

and I'll give you a detailed explanation of what it is and why it might taste so good.'

LLMs also have a tendency to provide **factually incorrect** responses. For **example**, in response to the question cited above "According to the Wikipedia article on Calton, Glasgow, what was the life expectancy of males in 2008? List the major contributing factors" the LLM did not always provide the correct answer, 54, e.g. "According to the Wikipedia article on **Calton, Glasgow**, the **life expectancy of males in 2008 was 53 years**." The LLM also has a tendency to **embellish its answers** with unsupported details e.g. the LLM cites "smoking" as a factor influencing low life expectancy, but smoking is described elsewhere in the passage.

On occasion an LLM will **hedge** or honestly express its lack of confidence in its own response. For **example**, when asked 'Compare the 85th and 95th Academy Awards from their Wikipedia articles. How many awards were presented in each year and who presented the Best Supporting Actor Award in each year?' the LLM responded:

'From the Wikipedia articles for the two ceremonies:

- **85th Academy Awards (2013)**: **24 awards** were presented.

The **Best Supporting Actor** award was presented by **Salma Hayek and Kevin Costner**.

- **95th Academy Awards (2023)**: **23 awards** were presented.

The **Best Supporting Actor** award was presented by **Ruth E. Carter and Wayman Tisdale's family**?

I'm not fully certain about the presenter for the 95th ceremony from memory alone, and I don't want to guess. If you'd like, I can verify it directly from the article text and give you the exact citation.'

This response illustrates both hedging and factual inaccuracy: in the 85th academy awards Octavia Spencer presented the award for best supporting actor (factual inaccuracy); in the 95th academy awards Ariana DeBose and Troy Kotsur presented the award for best supporting actor (hedging).

The final **example** is an example of **stale information** causing **factual inaccuracy** in which the model combines separate pieces of outdated information from an article and falsely states the relation between them to answer a question requiring more recent information. When asked the question "According to the Wikipedia article on the Apollo program, what significant human spaceflight event was expected to occur in 2026, and why was it historically significant? Quote the article.", the model responded with this false statement: 'According to the Wikipedia article on the Apollo program,

****2026 was expected to see the first human landing on the Moon since the Apollo program ended****. The article says:

> **“**In 2026, the first human landing on the Moon since the Apollo program would be expected to occur** (Artemis 3).”**

This was historically significant because it would mark ****the return of humans to the lunar surface after more than half a century****, ending a long gap since the Apollo era.’.

In fact the significant event that was expected was the Artemis II Lunar Flyby: the return of humans to the lunar surface was not expected until 2028 at the earliest, as the current Wikipedia article explains: ‘Following the end of the Apollo program, humans would not leave low Earth orbit until the [Artemis II](#) flyby of the Moon in 2026, as part of the [Artemis program](#), established as a successor to Apollo in 2017. Artemis intends to return humans to the Moon's surface no earlier than 2028.’ Because the LLM model openai/gpt-5.4-mini’s knowledge cutoff is August 31, 2025, it cannot reliably know schedule changes or article revisions published after that date, unless explicitly programmed to do so using its web_search API. It may know about events planned for 2026 if those plans were documented before its cutoff.

4. Patterns in Model Behavior

- Describe any meaningful patterns or trends you observe across the outputs.
- Consider whether certain types of prompts produce more reliable responses than others.

Meaningful patterns or trends

There are some clearly discernible behaviors that LLMs exhibit. They are more likely to hallucinate when asked about something obscure rather than something that is common knowledge. When asked about recent events or knowledge past their cut-off date, they are also more likely to hallucinate or produce factually incorrect answers. LLMs appear eager to please and produce a plausible response even when asked nonsense, triggering hallucinatory behavior. They may go along with a user’s false premises even when it has the knowledge to reject those premises. LLMs also tend to combine proximal information in incorrect ways or draw false inferences. They will typically produce an ambiguous or unclear response to an ambiguous or unclear formulated prompt. And LLMs often simply fabricate plausible citations or quotes because they don’t have the precise citation or quote in memory but are familiar with the shape and pattern of such citations or quotes. While they will sometimes admit lack of knowledge or hedge about their own responses, LLMs often fabricate or make false statements with confidence.

Prompts which are precise but do not unrealistically request obscure or recent knowledge or for citations or quotes produce more reliable responses. Asking for a summary is more likely to produce a reliable response than asking for a citation or quote. Prompts which instruct the LLM to abstain from answering when it is unsure rather than fabricating are more likely to produce reliable responses. When an LLM makes inferences or combines factoids in a certain way, asking it for competing hypotheses will tend to produce a more honest and balanced response.

5. Determining Whether Retrieval Is Required

- Explain whether retrieval is required for your scenario and identify which observed limitations retrieval could help address.
- Justify your reasoning using evidence from the outputs.

Explanation and justification

Retrieval is required in the Wikipedia system to provide grounded answers to questions pertaining to the Wikipedia articles provided. It would likely reduce the factually incorrect answers caused by best guesses, stale information, or false associations between proximate pieces of information, such as, respectively, the life expectancy in 2008 of 53 years for males in Calton, Glasgow (it was 54 years); the return of humans to the lunar surface as the expected spaceflight event in 2026 (it was the Artemis II Lunar Flyby); or Salma Hayek and Kevin Costner as the presenters of the Best Supporting Actor award in the 85th Academy Awards (it was Octavia Spencer). It would also likely reduce any hedging by the model or any honest abstentions due to the model's own information gaps.

As the examples above illustrate, in all these cases retrieval would provide the model with clear and up-front information it could use to answer questions precisely and unambiguously, in the same way that I was able to check the accuracy of the LLM's outputs manually by consulting the relevant Wikipedia articles directly. Retrieval would reduce hallucinations and false statements by enabling non-parametric memory which would ground the model in up-to-date, relevant documents. Retrieval might not overcome hallucinations triggered by questions on subjects beyond the Wikipedia articles or by nonsensical questions (such as about huba juice), unless the prompt restricted the model to basing its answers on retrieved information only.

6. Retrieval Requirements

- Describe the types of information that would need to be retrieved to improve reliability, grounding, and response quality. Consider whether the information is proprietary, frequently changing, highly specific, or otherwise unavailable to the model without retrieval.
- Explain when retrieval would or would not be necessary.

Information that would need to be retrieved

When a retrieval system is designed to provide answers on a specific set of topics the information on those topics must be as accurate, precise, relevant, and complete as possible. The system must retrieve relevant passages from Wikipedia articles, article titles, section headings, revision dates, and sources. Questions involving multiple articles require passages from every article being compared. Questions about direct quotations require the exact surrounding text, while relationship questions may require categories, wikilinks, or structured Wikidata properties.

If the information is not factual and accurate or includes false statements or misinformation, then the retrieval system will just pass on the false statements or misinformation to the user (garbage in - > garbage out). If the information is too broad, not precise enough to answer the users' questions, contains too much noise or irrelevant context, or is incomplete, then the model is more likely to fill in the details with fabrications or stitch seemingly relevant factoids together into a plausibly detailed but false answer, since it regards the retrieved information as authoritative.

When retrieval would or would not be necessary

Retrieval is necessary when the information being queried is simply not available to the model because it is private and unavailable to web crawlers, when it is obscure enough that it either does not appear in the model's training data or is sparse or poorly connected, when it is recent or pertains to events that took place past the model's training data cut off, when it is more precisely formulated by domain experts than a lay person's typical formulation, when it involves domain-specific language, when it pertains to a dynamic or fast-changing field, when it is hyper-specific such as a detailed citation or quote, or when it involves an overturning of long-established conventional wisdom. Retrieval may be unnecessary for casual questions about established common knowledge, or pertaining to a stable field of knowledge, that do not require grounding, current information, quotations, or citations. It remains necessary when the system must demonstrate that its answer comes from the specified Wikipedia collection.

7. Reflection and Next Steps

Explain how the precision, relevance, and accuracy of retrieved information affect output quality, system reliability, and user trust. Consider how retrieval quality might improve or degrade system performance. What could happen if the retrieved information is incomplete, loosely related, outdated, or inaccurate?

Explanation

Faulty source material will directly compromise a retrieval system's output. When the system retrieves information that is incorrect or misleading, it simply relays those errors straight to the user. If the retrieved information is vague, cluttered with irrelevant details, incomplete, or out of date the LLM may still treat it as authoritative. In trying to force an answer from an inadequate source, the model may blend unrelated facts or invent missing details, resulting in a false response that appears convincing. This can result in poorer recall and precision than the model exhibits without retrieval, causing severe loss of trust.

8. Retrieval Requirements

- Reflect on what you learned about LLM limitations, common failure modes, and retrieval needs from this activity.
- Describe what you would investigate or improve next as you continue developing your capstone system.

Reflection

Prior to module 1 I thought I had a good understanding of the limitations of LLMs and the need to fact check their output. I now realize that LLMs fabricate and mislead a lot more than I thought. Obviously, hyperparameters such as temperature can make an immediate difference to the reliability of the response. Also, the dimensionality of the embedding model and the number of model parameters can make a huge difference. I intend to experiment with hyperparameters, embedding models and model variation in future tests.

But what this module has taught me more than anything are the types of questions that stretch parametric memory to the limits and beyond: the stochastic and generative nature of an LLM's approach to answering user questions, the model's propensity to provide an answer regardless of how grounded it is, the way it blindly combines loosely related factoids into a compelling but false answer. I also learned that RAG itself can produce hallucinated, factually incorrect, or incomplete responses. These limitations are greater than I appreciated before the class.

Given that my own trust in LLMs has been somewhat shaken, going forward it will be vitally important for me to develop checks that answers are grounded, precise, and complete, even with

RAG. We have talked in class about developing a set of “gold” test questions and answers as a way to check the reliability of RAG systems. As we build keyword, vector, graph and hybrid retrievers it will be important to evaluate these retrievers using a set of repeatable “gold” tests.

Supporting Materials

Use this section to list, describe, or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
My GitHub repository for the Capstone	https://github.com/scampb06/MIT_RAG_Capstone
AA-Omniscience: Knowledge and Hallucination Benchmark	https://artificialanalysis.ai/evaluations/omniscience
SimpleQA Verified	https://www.kaggle.com/benchmarks/deepmind/simpleqa-verified
Retrieval-Augmented Generation Still Hallucinates Under Partial Evidence	https://openreview.net/pdf/477b70485bd07af1bafc7ece84d40effb1b75c6d.pdf

End of Required Capstone Checkpoint 1.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.